KING SALMAN INTERNATIONAL UNIVERSITY

Field of Engineering and Computer Engineering

All Program

Course : Mathematics 1 (MAT 111)

Lecture 4: Limits and Continuity

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Course contents

Calculus part

- Functions
- Limits
- Continuity
- Derivatives
- Applications of differentiation

Linear Algebra part

- Systems of Linear Equations
- Theory of Matrices
- Applications in Systems of Equations

textbook: R. Smith, R. Minton, z. Rafhi, Calculus: Early Transcendental Functions, 5th Edition, McGraw-Hill Education.

Limits

Definition: the function f x approaches L when x approaches a . In this case we say that f x has a limit L when x tends to a and written as $\lim_{x\to a} f(x) = L$

Example 1: find the limit of

i. $\lim_{x\to 1} 2x^2 + 3x + 1 = 2(1) + 3(1) + 1 = 6$

ii. $\lim_{x\to 1} \frac{x^2+1}{x+1} = \frac{1+1}{1+1} = 1$

iii. $\lim_{x\to 0} \frac{\sin x}{1+\cos x} = \frac{0}{2} = 0$

iv. $\lim_{x\to 1} \frac{x^2+1}{\log x} = \frac{2}{0} = \infty$

one-sided-limit
Right and Left Limit

The Right limit means that the limit when x closes to the number a or the number greater than it and can be written as

$$\lim_{x \to a^+} f(x)$$

The Left limit means that the limit when x closes to the number a or the number less than it and can be written as

$$\lim_{x \to a^-} f(x)$$

Example 2: find the right and left limit of

$$f(x) = \frac{x}{|x|} \; when \; x \to 0$$

Solution: rewrite the function as

$$f(x) = \frac{x}{|x|} = \begin{cases} \frac{x}{x}, & x \geq 0 \\ \frac{x}{-x}, & x < 0 \end{cases} = \begin{cases} 1, & x \geq 0 \\ -1, & x < 0 \end{cases}$$

The right limit is

$$\lim_{x \to 0^+} f(x) = 1$$

and the left limit is

$$\lim_{x \to 0^-} f(x) = -1$$

Theorem 1

Theorem 1: the function $f(x)$ has a limit if its right limit and its left limit are exist and equal.

i.e.

$$\lim_{x \to a} f(x) = L$$

If

$$\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = L$$

But if

$$\lim_{x \to a^+} f(x) \neq \lim_{x \to a^-} f(x)$$

Then $\lim_{x \to a} f(x)$ doesn't exist.

Example 3: discuss the limit

$$\lim_{x\to 1}|x|$$

Solution:

$$\lim_{x\to 1}|x| = \lim_{x\to 1}\begin{cases} x, & x \geq 1 \\ -x, & x < 1 \end{cases}$$

then

$$\lim_{x\to 1^+}|x| = \lim_{x\to 1^+} x = 1$$

And

$$\lim_{x\to 1^-}|x| = \lim_{x\to 1^-}(-x) = -1$$

Since $\lim_{x\to 1^+}|x| \neq \lim_{x\to 1^-}|x|$ then $\lim_{x\to 1}|x|$ not exist.

Algebra of limits

Theorem 2: given $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$ then the following is true

- $\lim_{x\to a}[f(x) \pm g(x)] = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x) = L \pm M$
- $\lim_{x\to a}[f(x) * g(x)] = \lim_{x\to a} f(x) * \lim_{x\to a} g(x) = L * M$
- $\lim_{x\to a}[\frac{f(x)}{g(x)}] = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)} = \frac{L}{M}$
- $\lim_{x\to a}(f(x)^n = (\lim_{x\to a} f(x))^n = L^n$
- $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)} = \sqrt[n]{L}$

Example 4: given $f(x) = x^2$ and $g(x) = \frac{4}{x+1}$ find

- $\lim_{x\to 1}[f(x) + g(x)] = \lim_{x\to 1} f(x) + \lim_{x\to 1} g(x) = \lim_{x\to 1} x^2 + \lim_{x\to 1} \frac{4}{x+1} = 1 + 2 = 3$
- $\lim_{x\to 1}[f(x) - g(x)] = \lim_{x\to 1} f(x) - \lim_{x\to 1} g(x) = \lim_{x\to 1} x^2 - \lim_{x\to 1} \frac{4}{x+1} = 1 - 2 = -1$
- $\lim_{x\to 1}[f(x).g(x)] = \lim_{x\to 1} f(x) . \lim_{x\to 1} g(x) = \lim_{x\to 1} x^2 . \lim_{x\to 1} \frac{4}{x+1} = 1 * 2 = 2$
- $\lim_{x\to 1}\left[\frac{f(x)}{g(x)}\right] = \frac{\lim_{x\to 1} f(x)}{\lim_{x\to 1} g(x)} = \frac{1}{2}$
- $\lim_{x\to 1}(g(x))^5 = (\lim_{x\to 1} g(x))^5 = 2^5 = 32$
- $\lim_{x\to 1} \sqrt[4]{f(x)} = \sqrt[4]{\lim_{x\to 1} f(x)} = \sqrt[4]{1} = 1$

Methods for Calculating the limits

(1) Direct substitution

To find the limit

$$\lim_{x \to a} f(x) = f(a)$$

Replace x by a.

This method used when $f(x)$ is polynomial, root function and rational function such that a be in the domain of f(x).

Example 5: find the following limits.

- $\lim_{x \to 6} \sqrt{x-2} = \sqrt{6-2} = \sqrt{4} = 2$
- $\lim_{x \to 0} x^5 + 1 = 0 + 1 = 1$
- $\lim_{x \to 3} \frac{x^3+1}{x-2} = \frac{2}{3-2} = 2$
- $\lim_{x \to 2} \frac{x^2-x-2}{x-2} = \frac{4-2-2}{2-2} = \frac{0}{0}$ undefined value كمية غير معرفة

(2) Canceling (deleting) zeros of denominator

If

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\frac{0}{0}\quad \text{undefined}$$

Then

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{k(x)g(x)}{g(x)}=\lim_{x\to a}k(x)=k(a)$$

نحلل البسط $(f(x))$ الى قوسين بحيث يكون المقام واحد منهم.

نتخلص من اصفار المقام ثم نستخدم التعويض المباشر.

فى بعض المسائل نحتاج نحلل المقام ايضا الى قوسين مثل البسط ثم نلغى القوس الذى يعطى صفرمن المقام والبسط.

Example 6: find the following limits

- $\lim_{x\to 2}\frac{x^2-x-2}{x-2}=\frac{4-2-2}{2-2}=\frac{0}{0}$ undefined value كمية غير معرفة then

$$\lim_{x\to 2}\frac{x^2-x-2}{x-2}=\lim_{x\to 2}\frac{(x-2)(x+1)}{x-2}=\lim_{x\to 2}(x+1)=3$$

- $\lim_{x\to 1}\frac{x^2+x-2}{x^2-x}=\lim_{x\to 1}\frac{x^2+x-2}{x(x-1)}=\frac{1+1-2}{1*(1-1)}=\frac{0}{0}$ then

$$\lim_{x\to 1}\frac{x^2+x-2}{x(x-1)}=\lim_{x\to 1}\frac{(x+2)(x-1)}{x(x-1)}=\lim_{x\to 1}\frac{(x+2)}{x}=3$$

- $\lim_{x\to 5}\frac{x^2-25}{x-5}=\frac{25-25}{5-5}=\frac{0}{0}$ then

$$\lim_{x\to 5}\frac{x^2-25}{x-5}=\lim_{x\to 5}\frac{(x-5)(x+5)}{x-5}=\lim_{x\to 5}(x+5)=10$$

- $\lim_{x\to 1}\frac{x^2+x-2}{x^2-1}=\frac{0}{0}$ then $\lim_{x\to 1}\frac{x^2+x-2}{x^2-1}=\lim_{x\to 1}\frac{(x+2)(x-1)}{(x+1)(x-1)}=\lim_{x\to 1}\frac{(x+2)}{(x+1)}=\frac{3}{2}$

Limits of Trigonometric Functions

- $\lim_{x\to 0}\frac{\sin x}{x}=1,$ $\lim_{x\to 0}\frac{x}{\sin x}=1$
  - $\lim_{x\to 0}\frac{\sin ax}{x}=a,$ $\lim_{x\to 0}\frac{x}{\sin ax}=\frac{1}{a}$
  - $\lim_{x\to 0}\frac{\sin x}{bx}=\frac{1}{b}$
  - $\lim_{x\to 0}\frac{\sin ax}{bx}=\frac{a}{b}$
- $\lim_{x\to 0}\frac{\tan x}{x}=1,$ $\lim_{x\to 0}\frac{x}{\tan x}=1$
  - $\lim_{x\to 0}\frac{\tan ax}{x}=a,$ $\lim_{x\to 0}\frac{x}{\tan ax}=\frac{1}{a}$
  - $\lim_{x\to 0}\frac{\tan x}{bx}=\frac{1}{b}$
  - $\lim_{x\to 0}\frac{\tan ax}{bx}=\frac{a}{b}$
- $\lim_{x\to 0}\frac{\sin ax}{\tan bx}=\frac{a}{b},$ $\lim_{x\to 0}\frac{\tan ax}{\sin bx}=\frac{a}{b}$

Example 7: find the following limits

- $\lim_{x\to\pi}(\sin x - 3x\cos x) = (0 - 3 * \pi * -1) = 3\pi$
- $\lim_{x\to 0}\frac{\sin 2x}{x} = \lim_{x\to 0}\frac{\sin 2x}{2x} * 2 = 1 * 2 = 2$
- $\lim_{x\to 0}\frac{\sin 2x}{\tan 3x} = \frac{2}{3}$
- $\lim_{x\to 0}\frac{\cos x}{1-\sin x} = \frac{1}{1-0} = 1$
- $\lim_{x\to 0}\frac{\sin 2x}{3x} = \frac{1}{3}\lim_{x\to 0}\frac{\sin 2x}{2x} * 2 = \frac{2}{3}$
- $\lim_{x\to 0}\frac{\tan x}{2x} = \frac{1}{2}$
- $\lim_{x\to 0}\frac{x}{x+\tan x}$ divide by x then

$$\lim_{x\to 0}\frac{1}{1+\frac{\tan x}{x}} = \frac{1}{1+\lim_{x\to 0}\frac{\tan x}{x}} = \frac{1}{1+1} = \frac{1}{2}$$

Limits when $x \to \infty$

$$\lim_{x\to\infty} f(x) = F(\infty)$$

Example (8): find the limit

- $\lim_{x\to\infty} (x^2 + 3) = (\infty + 3) = \infty$
- $\lim_{x\to\infty} \frac{1}{x+3} = \frac{1}{\infty} = 0$
- $\lim_{x\to\infty} \frac{x+6}{x^2+3} = \frac{\infty}{\infty}$ undefined كمية غير معرفة

Remark 1: if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$

نقسم على اكبر اس ل x فى البسط والمقام

So in the last example $\lim_{x\to\infty} \frac{x+6}{x^2+3} = \frac{\infty}{\infty}$ divide by x^2 then

$$\lim_{x\to\infty} \frac{x+6}{x^2+3} = \lim_{x\to\infty} \frac{\frac{1}{x} + \frac{6}{x^2}}{1 + \frac{3}{x^2}} = \frac{0+0}{1+0} = \frac{0}{1} = 0$$

Example 9: find the following limits

- $\lim_{x\to\infty}\frac{2x+3}{x-3}=\frac{\infty}{\infty}$ divide by x then $\lim_{x\to\infty}\frac{2x+3}{x-3}=\lim_{x\to\infty}\frac{2+\frac{3}{\infty}}{1-\frac{3}{\infty}}=\frac{2+0}{1-0}=2$
- $\lim_{x\to\infty}\frac{4x^3-2x}{x^2+3x-5}=\frac{\infty}{\infty}$ divide by x^3 then

$$\lim_{x\to\infty}\frac{4x^3-2x}{x^2+3x-5}=\lim_{x\to\infty}\frac{4-\frac{2}{x^2}}{\frac{1}{x}+\frac{3}{x^2}-\frac{5}{x^3}}=\frac{4-0}{0}=\frac{4}{0}=\infty$$

- $\lim_{x\to\infty}\frac{x+7}{x^2+2x}=\frac{\infty}{\infty}$ divide by x^2 then $\lim_{x\to\infty}\frac{x+7}{x^2+2x}=\lim_{x\to\infty}\frac{\frac{1}{x}+\frac{7}{x^2}}{1+\frac{2}{x}}=\frac{0}{1}=0$
- $\lim_{x\to\infty}\sqrt{\frac{x^2+2}{x^2+1}}=\frac{\infty}{\infty}$ divide by x then $\lim_{x\to\infty}\sqrt{\frac{x^2+2}{x^2+1}}=\lim_{x\to\infty}\sqrt{\frac{1+\frac{2}{x^2}}{1+\frac{1}{x^2}}}=\sqrt{\frac{1+0}{1+0}}=1$

Continuity of Functions اتصال الدوال

Def.: the function $f(x)$ is continuous at the point $x = a$ if it satisfy the following conditions

- $f(a)$is defined (exist)
- $\lim_{x \to a} f(x)$ exist i.e. $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x)$
- $\lim_{x \to a} f(x) = f(a)$

If one of the above conditions not satisfied then $f(x)$ is called discontinuous

Example 10: discuss the continuity of the following functions

- $f(x) = x^2$ since x^2 is polynomial then it is defined for values of $x \in (-\infty, \infty)$
  so x^2 is continuous on $(-\infty, \infty)$.
- $f(x) = \sin x$ since $\sin x$ is defined for values of $x \in (-\infty, \infty)$
  so $\sin x$ is continuous on $(-\infty, \infty)$.
- $f(x) = \frac{x^2}{x-1}$ since $\frac{x^2}{x-1}$ is defined for values of $x \in (-\infty, \infty)$ except at $x = 1$
  so $\frac{x^2}{x-1}$ is continuous on $(-\infty, \infty) - \{1\}$.
- $f(x) = \sqrt{x-2}$ since $\sqrt{x-2}$ is defined for values of $x \in [2, \infty)$
  so $\sqrt{x-2}$ is continuous on $[2, \infty)$.

Remark 2: From the previous example we see that the polynomial function, trigonometric functions, root functions and rational functions are continuous on their domains.

Example (11): discuss the continuity of the function

$$f(x)=\begin{cases}\frac{x^2-x-6}{x-3}, & x\neq 3\\ 5, & x=3\end{cases} \text{ at } x=3$$

Solution:

- $f(3)=5$
- $\lim\limits_{x\to 3^+} f(x)=\lim\limits_{x\to 3^-} f(x)=\lim\limits_{x\to 3}\frac{x^2-x-6}{x-3}$

$$=\lim_{x\to 3}\frac{(x-3)(x+2)}{x-3}=\lim_{x\to 3}(x+2)=5$$

- $\lim\limits_{x\to 3} f(x)=f(3)$

so $f(x)$ is continuous at $x=3$.

Example (12): discuss the continuity of the function

$$f(x)=\begin{cases} x+2, & x>1 \\ x^2, & x\le 1 \end{cases} \text{ at } x=1$$

Solution:

- $f(1)=1^2=1$
- $\lim\limits_{x\to 1^+} f(x)=\lim\limits_{x\to 1^+} x+2=3$

  $\lim\limits_{x\to 1^-} f(x)=\lim\limits_{x\to 1^-} x^2=1$

Then $\lim\limits_{x\to 1^+} f(x)\neq \lim\limits_{x\to 1^-} f(x)$ so $\lim\limits_{x\to 1} f(x)$ doesn't exist

so $f(x)$ is discontinuous at $x=1$.

Example (13): Find the value of k such that the following function becomes continuous

$$f(x) = \begin{cases} kx + 6, & x > 3 \\ kx^2, & x \leq 3 \end{cases}$$

Solution:

Since $f(x)$ is continuous at any point so it is continuous at $x = 3$.

So $f(3) = \lim_{x \to 3} f(x)$

$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} kx + 6 = 3k + 6$$

$$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} kx^2 = 9k$$

Also $f(3) = 9k$ then $f(3) = \lim_{x \to 3} f(x) \to 3k + 6 = 9k \to \boxed{k = 1}$

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THANK YOU

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